

Information Communication Technologies

SotM EU
VIENNA 15-17 JULY 2011

Prototype of a Mobile Application for the Improvement of OSM Data Quality

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salzburg research

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Motivation









Research Objectives

- a framework for mobile data acquisition and improvement
- for different domains
- as simple as possible
- for GPS-based smartphones



Content

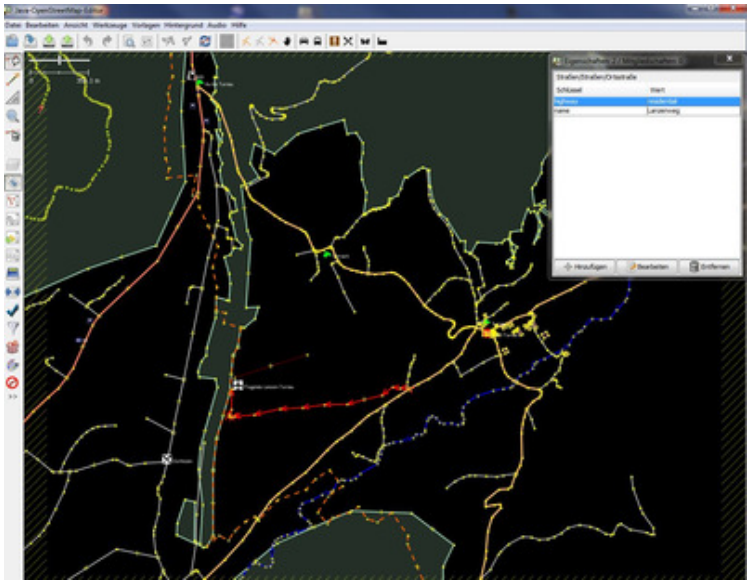
- Why Mobile Data Improvement for OSM?
- Requirements
- Prototype
- First User Tests
- Future Work





Why Mobile Data Improvement for OSM?

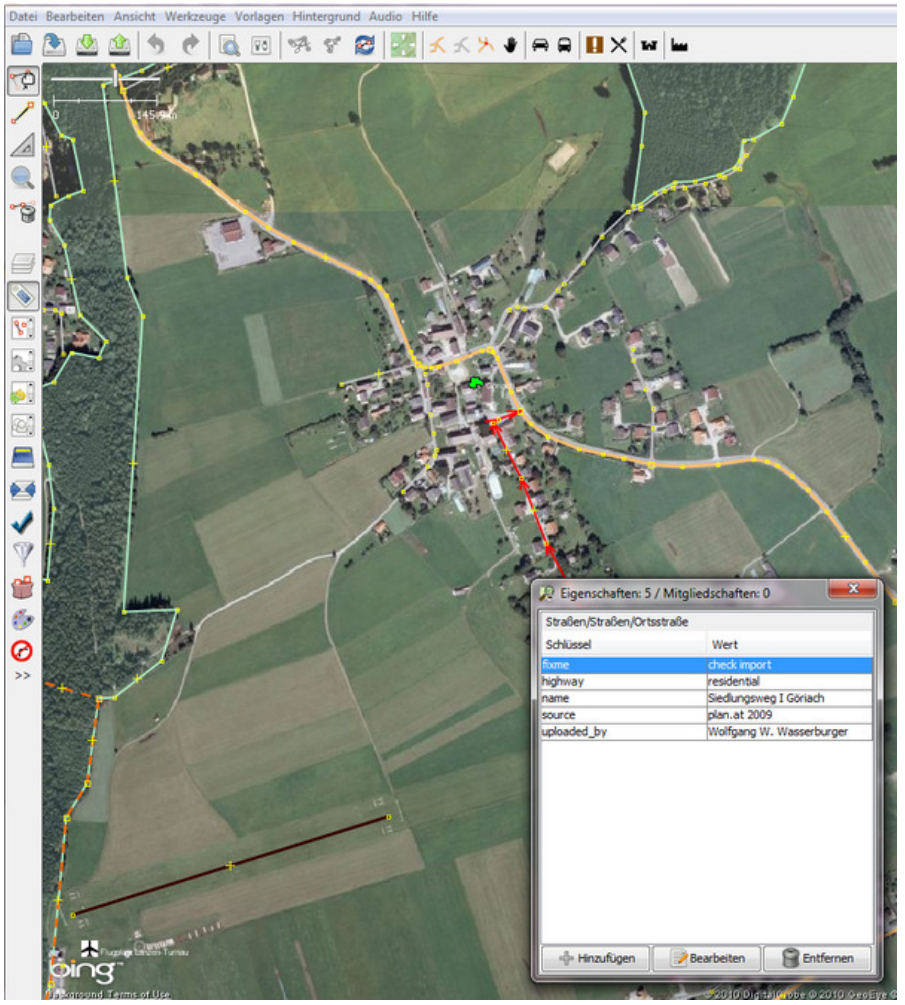
- quantity of data vs. quality of data
  - urban areas vs. rural areas
  - white places vs. detailed map features
- tools for experts
- new applications demonstrate missing tags





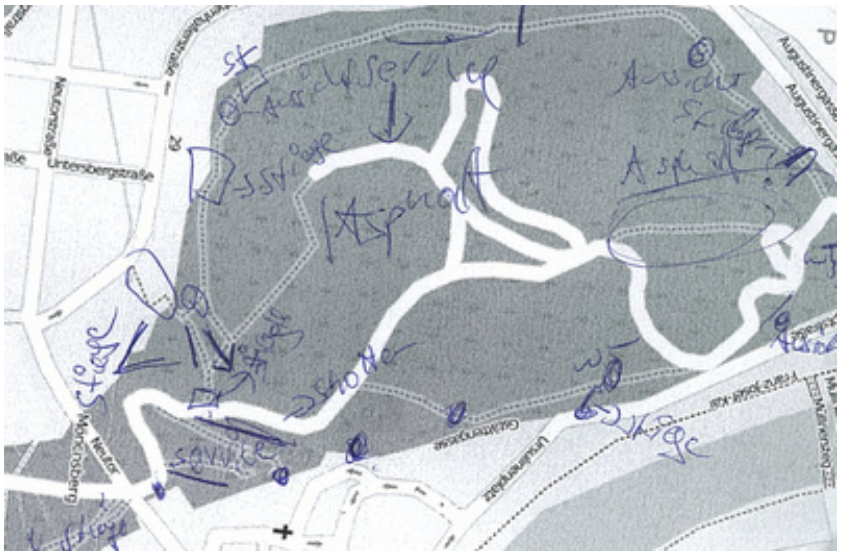
Data Acquisition with Satellite and Aerial Images

- data acquisition at home
- good for basic data structures
  - streets
  - buildings
  - land cover
- no detailed map features



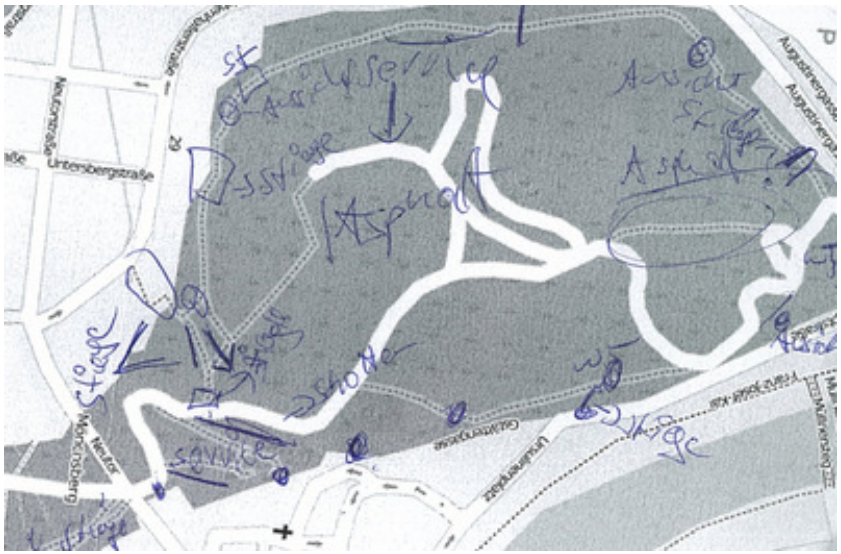
Data Acquisition and Improvement in the Field

- GPS-Logger, digital photos, pencil and paper



Data Acquisition and Improvement in the Field

- GPS-Logger, digital photos, pencil and paper



- smartphone apps for data improvement



Smartphone Apps for OSM (OSM-Wiki 08-07-2011)

| Android

OpenStreetMap editing features

Name	Add POIs	Edit / Delete POIs	Edit arbitrary tags of existing OSM objects	Edit geometries	Edit relations	Upload to OSM
GpsMid	Yes	yes (starting from nightly 2011-05-30)	Yes	No	No	Yes
KeypadMapper	?	?	?	?	?	?
Mapzen POI Collector	Yes	Yes	Yes	No	No	Yes
OSMTracker for Android	Yes	?	?	?	?	?
Vespucci	Yes	Yes	Yes	Yes	No	Yes

| iPhone

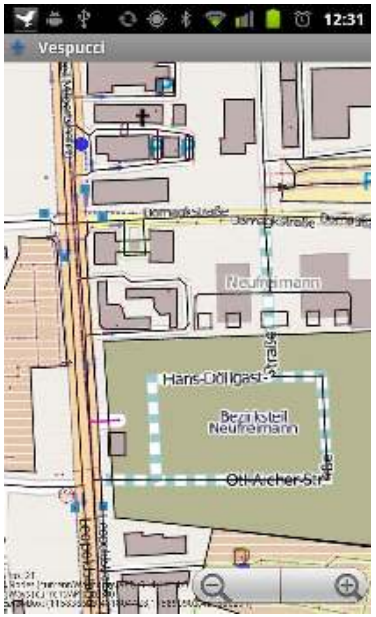
OpenStreetMap editing features

Name	Add POIs	Edit / Delete POIs	Edit arbitrary tags of existing OSM objects	Edit geometries	Edit relations	Upload to OSM
Bluemapia	Yes	?	?	?	?	Yes
Cycle Hire	?	?	?	?	?	?
iLOE	Yes	Yes	Yes	No	No	Yes
Mapzen POI Collector	Yes	Yes	Yes	No	No	Yes
Osmeditor	?	?	?	?	?	?


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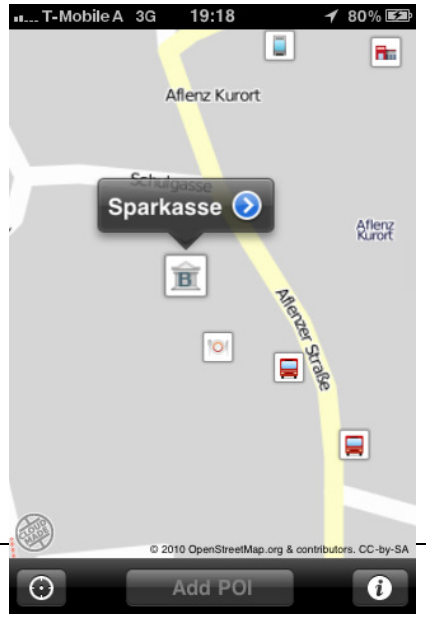

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Bezirksteil Neufriemann
Ott-Archer-Straße


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© 2010 OpenStreetMap.org & contributors, CC-by-SA
Add POI







Information Communication Technologies



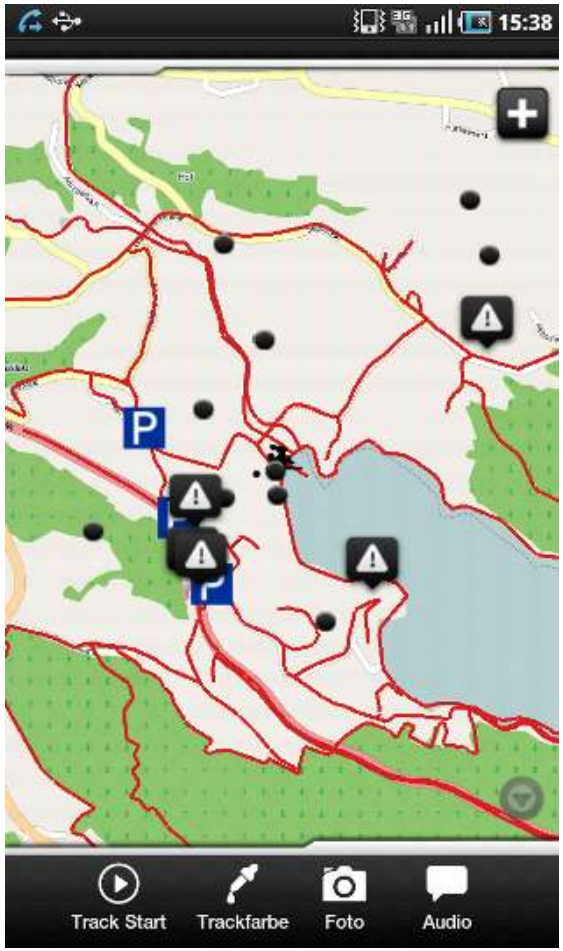
Requirements

- a app for everybody
- user interface for small touchable displays
- in addition to desktop based editors
  - no functions for editing/deleting features
  - functions for the data improvement
- visualisation of missing tags in the OSM data set
- use of map features from the OSM-Wiki



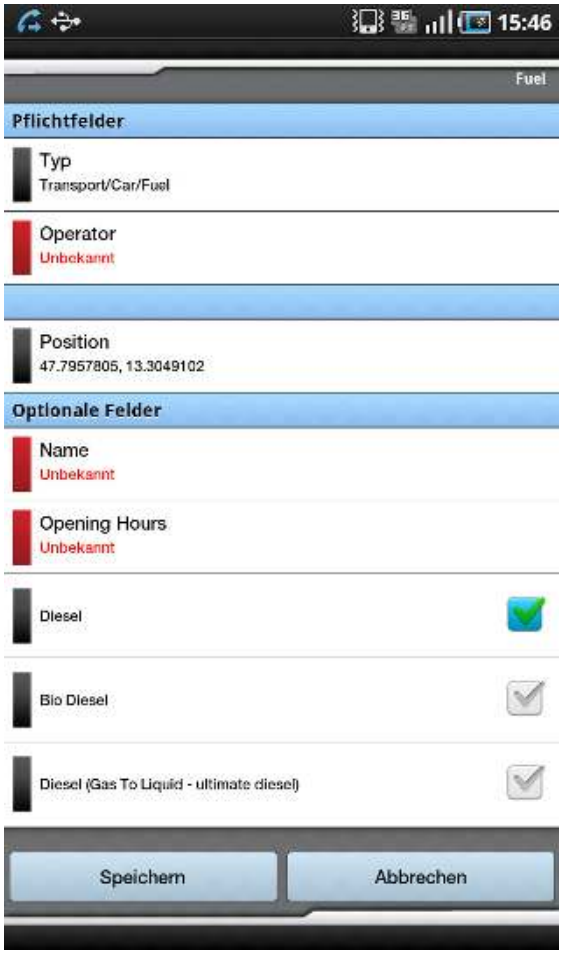
Prototype

- for Android smartphones
- central element is the map interface
- optional functions for
  - track recording
  - pictures and voice memos
- visualisation of features with missing tags

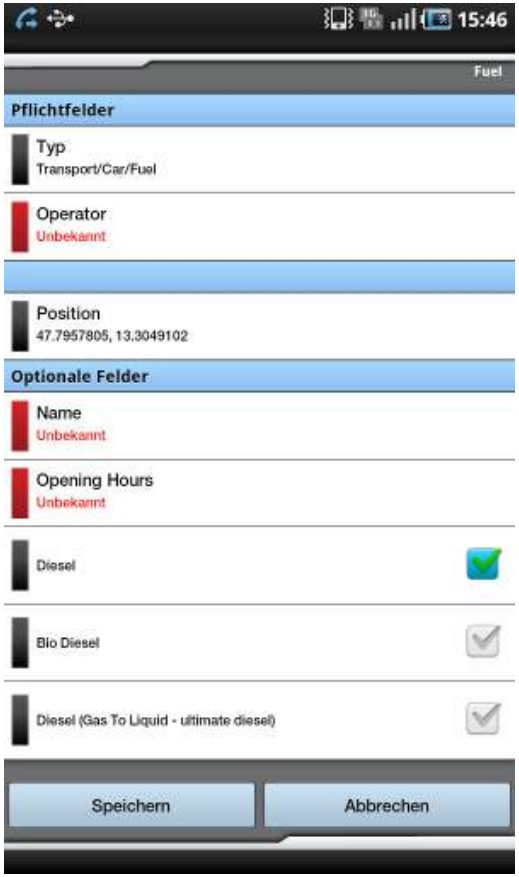


Automatic Detection of Missing Tags

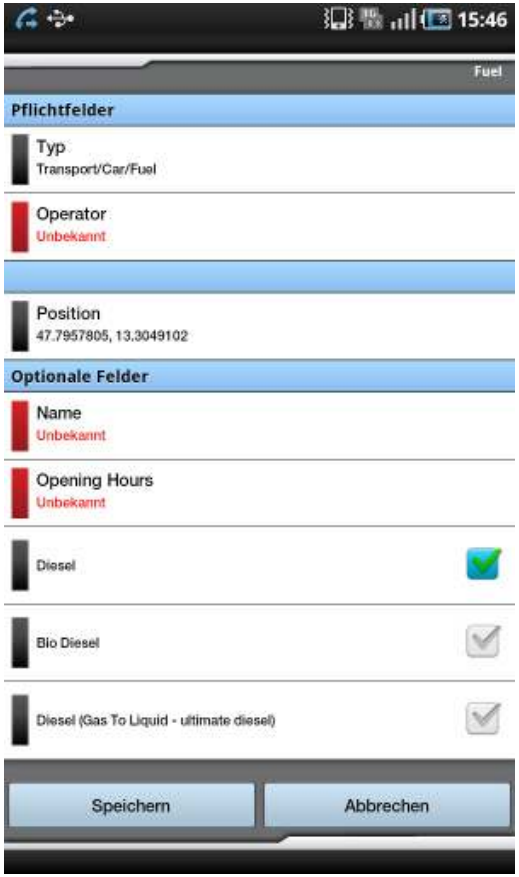
- map features
  - POI's
  - streets
- detection of completeness of features
  - definition of an ontology with the default preset from JOSM
  - classification in mandatory and optional tags

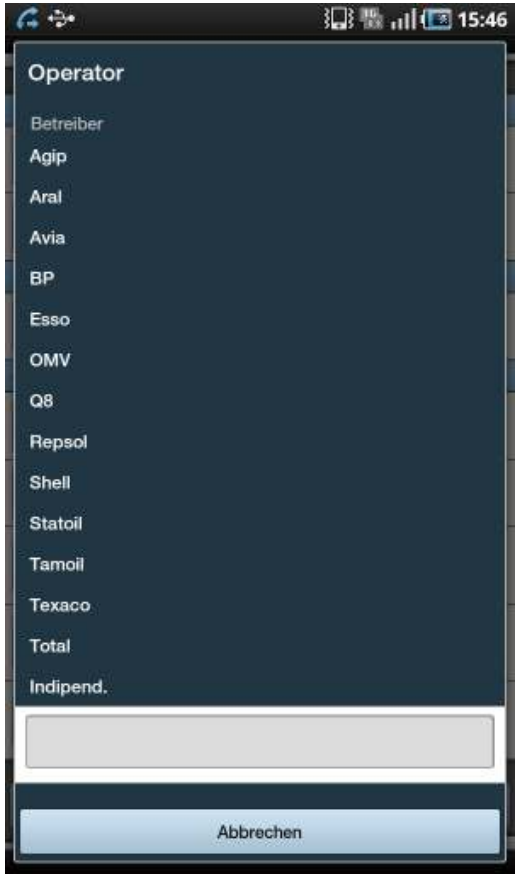


Improvement of Point-of-Interest

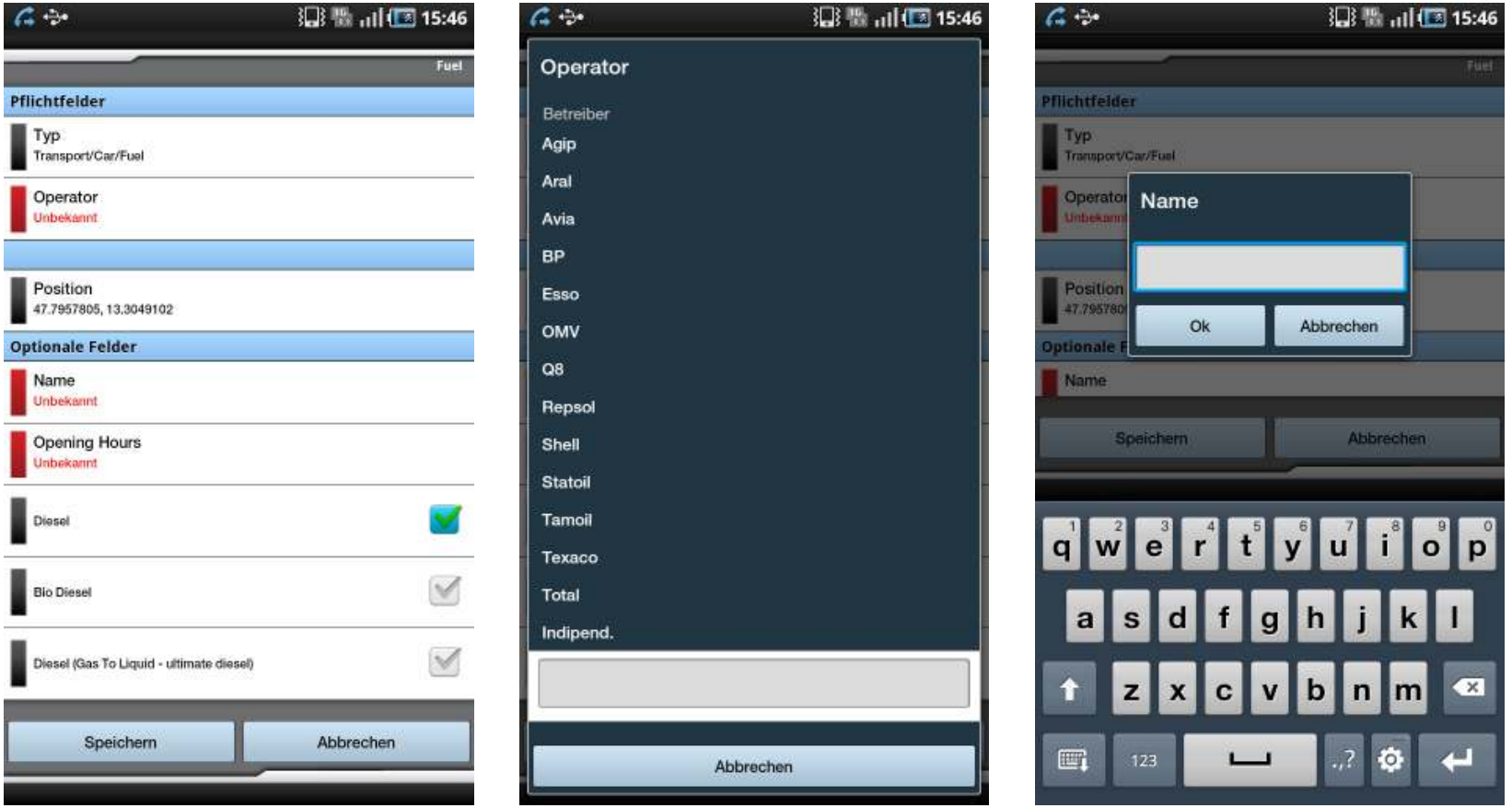


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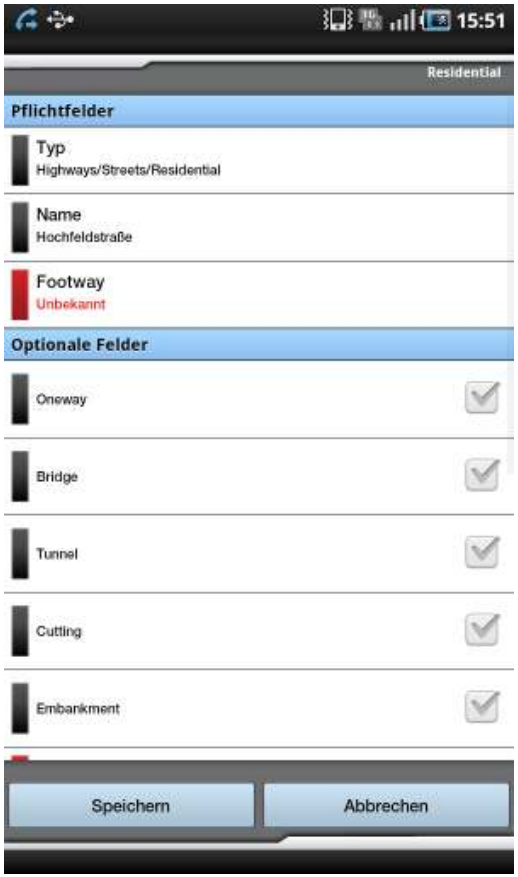


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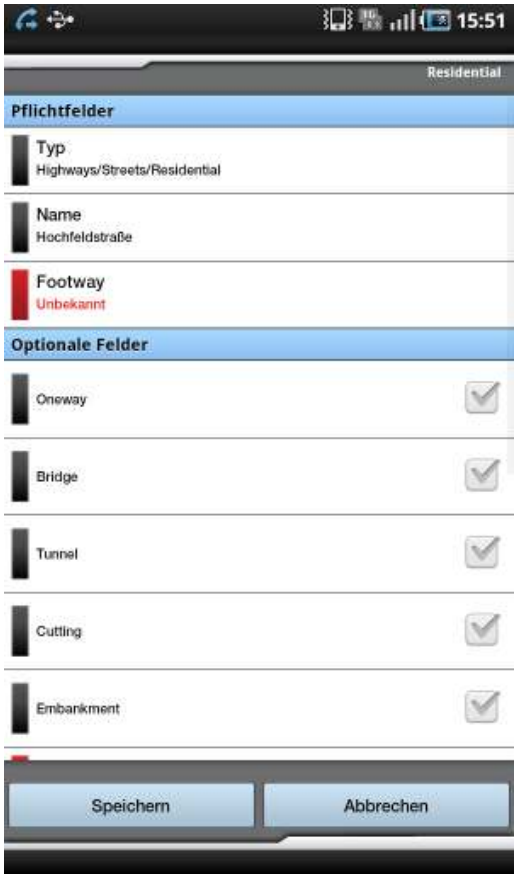


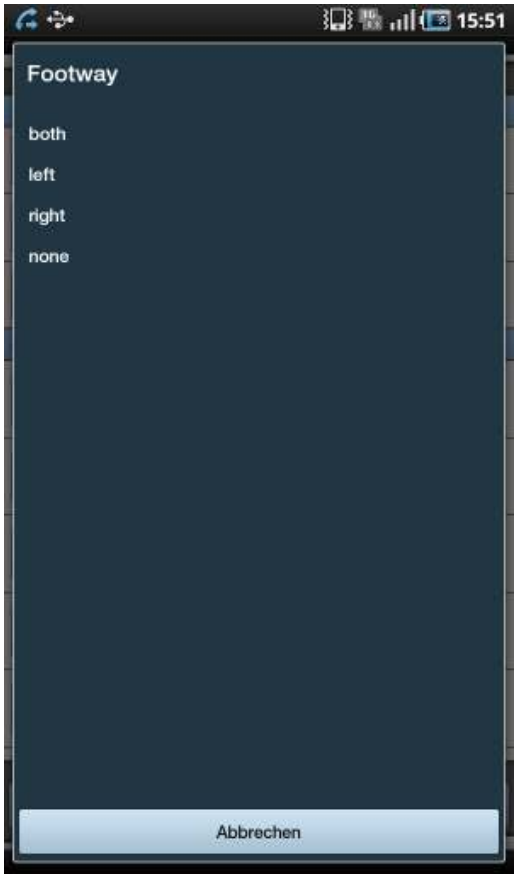


Improvement of Street Segments



Improvement of Street Segments







Feedback from User Tests

- tests with persons who have no experience with editing OpenStreetMap



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- Positive
  - *The usability of the user interface is very simple, practical and intuitive.*
  - *It is good, that there are predefined tags at the different map features.*



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- Constructive
  - *The zoom levels in the mobile application are too low.*
  - *The visualisation of many nearby POI's with missing tags is problematic.*



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- Constructive
  - *The zoom levels in the mobile application are too low.*
  - *The visualisation of many nearby POI's with missing tags is problematic.*

- Negative
  - *Sometimes the application is very slow.*



Future Work

- user interaction with touch displays, for example
  - tools for data acquisition
  - gestures for collecting emotional parameters (EmoMap)



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  - tools for data acquisition
  - gestures for collecting emotional parameters (EmoMap)
- adapting the prototype for blind users



Future Work

- user interaction with touch displays, for example
  - tools for data acquisition
  - gestures for collecting emotional parameters (EmoMap)
- adapting the prototype for blind users
- an operable app for the improvement of OpenStreetMap data